

Aren Desai

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Education

Georgia Institute of Technology - <i>Online M.S. Computer Science (in progress), ML Specialization</i>	expected Summer 2029
University of Wisconsin - Madison - <i>B.S. Computer Science & Data Science</i>	Spring 2025
Google Cloud - <i>Professional Machine Learning Engineer Certification</i>	Spring 2025
FICO - <i>FICO DMP Certification</i>	Summer 2024

Technical Skills

Programming Languages: Python, SQL, R, Java, C

Tools & Platforms: Microsoft Fabric, Microsoft Azure, PowerBI, Sharepoint, Google Cloud Platform, PCI Energy Solutions, PLEXOS

Experience

Madison Gas & Electric - *Madison, WI* May 2024 – Present
Energy Supply Analyst

- Built MGE's first ML-based Locational Marginal Price (LMP) forecaster and deployed it to production; the model — a stacked-generalization approach combining a gradient-boosted regression tree with a foundation model — outperforms the prior rules-based forecast by 12.06% in back-testing (Jul 2024–Jul 2026) and retrains automatically each week as new data arrives.
- Developed a Resource Adequacy (RA) hour classification model producing rolling 120-hour probability forecasts, using a gradient-boosted decision tree with SMOTE to address class imbalance from the rarity of RA hours — supporting resource planning and risk assessment.
- Built a Market Simulator in Python (SciPy, OR-Tools) used daily by the energy trading and generation planning teams to evaluate scenarios and inform decisions affecting hundreds of MWh.
- Performed ongoing model validation, back-testing, and reasonableness checks, and communicated forecast methodology, assumptions, and limitations to traders, generation planners, and analysts.
- Led the build-out of Microsoft Fabric and MLflow as MGE's platform for deploying, monitoring, and documenting forecasting and ML models for LMPs.
- Maintained a large, business-critical PCI Beanshell codebase responsible for settlements and profit calculations across all energy generating units, relied on by generation operators, analysts, and company-wide reporting.
- Mentored an intern through capstone projects.

Energy Supply and Trading Intern

- Partnered with DevOps and SecOps engineers to evaluate and stand up infrastructure options (on-prem GPU, AWS, existing Linux VMs, Azure) for deploying ML models into MGE's environment, ultimately landing on Microsoft Azure.

Compeer Financial - *Sun Prairie, WI* Jan 2023 – Jan 2024
Data Analytics Intern

- Built a scikit-learn model for the risk analysis team to predict whether a loan was likely to be referred to the Risk Assessment Unit, supporting risk-based decision-making.
- Helped introduce PyTorch and other machine-learning tooling into Compeer's analytics tech stack.

Leadership Experience

Google Developer Student Club Madison, WI
Finance Lead / Treasurer Sep 2023 – May 2024

- Managed club finances with UW Credit Union; organized tech talks and workshops
- Judged the 2024 CheeseHacks hackathon